

THE TALKING TOOLBOX

VOLTAGE DROP

Global Sovereign University - Free forever. Works on any phone. No login.

THE FORMULA

$VD = 2 \times K \times I \times L / CM$ (three-phase: use 1.732)

K FACTORS

Copper 12.9 | Aluminum 21.2 (ohm-cmil/ft)

CIRCULAR MILS

14 AWG 4,110 | 12 AWG 6,530 | 10 AWG 10,380 | 8 AWG 16,510

6 AWG 26,240 | 4 AWG 41,740 | 2 AWG 66,360 | 1/0 105,600

2/0 133,100 | 4/0 211,600 | 250 kcmil 250,000 | 500 kcmil 500,000

THE LIMITS

NEC notes: branch $\leq 3\%$ | feeder + branch $\leq 5\%$



SCAN TO OPEN THE LIVE TOOL - ASK GENO TO WORK IT OUT LOUD

<https://read.globalsovereignuniversity.org/voltage-drop-calculator.html> | From the book Job-Site Math by Dr. Gene A. Constant | Verify against current code

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CONDUIT FILL

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FILL LIMITS (NEC CH. 9)

1 wire 53% | 2 wires 31% | 3+ wires 40% | nipple ≤ 24 in 60%

THHN AREAS (in²)

14: .0097 | 12: .0133 | 10: .0211 | 8: .0366 | 6: .0507

4: .0824 | 2: .1158 | 1/0: .1855 | 4/0: .3237

EMT AREAS (in²)

1/2: .304 | 3/4: .533 | 1: .864 | 1-1/4: 1.496

1-1/2: 2.036 | 2: 3.356 | 2-1/2: 5.858 | 3: 8.846 | 4: 14.753

THE METHOD

Total all conductor areas, divide by fill %, pick the pipe



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HVAC BTU SIZING

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BASE RULE

About 20 BTU per square foot of cooling

ADJUST

Very sunny +10% | shaded -10% | poor insulation +15%

tight build -10% | each occupant past two +600 | kitchen +4,000

TONS

1 ton = 12,000 BTU/hr (24,000 BTU mini-split = 2 tons)

THE SIN

Oversizing short-cycles: cold, clammy, compressor-killing

THE REAL STANDARD

Whole-home sizing = ACCA Manual J by a professional



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